Course Content:

#	Content	Duration	CLOs
Mid-Term (30 Marks)			
1	Computer architecture Basic, History, Different terminologies of computing device, Types of Computer Architecture Understanding Program Performance, Defining	**2 weeks**	CLO1

	Performance, Measuring Performance, CPU Performance and its factor, Evaluating performance, MIPS as a performance Measure		
2	Instruction and data access methods, Instruction Set, Stored-Program Concept, Operations of the computer Hardware, Operands of the computer Hardware (Design Principles of Computer Hardware) Representation of Instructions in the Computer, Logical Operations, Instructions for decision making, MIPS Addressing for 32-Bit Immediate and Addresses	**2 weeks**	CLO1, CLO2
3	Arithmetic and logical operations, floating point operations, ALU design, Signed and Unsigned numbers, Number Conversion and representation, Arithmetic Operations and Representation Matrix-chain multiplication and longest common subsequence problems as examples, Complexity analysis of the algorithms. Multiplication, Division and Floating point Hardware	**2 weeks**	CLO2
Final Exam: 50 Marks **Group-A (20 Marks)**			
4	The control unit (Single cycle Data path) : hardwired and micro programmed, Logic Design Convention, Clocking Methodology, Data path Basic, State Element of Data path Building a single Data path (R-Type, I-Type and J-type Instructions) Designing the main control unit (with control signals effect), Operation of the data path	**2 week**	CLO2
5	The control unit (Multiple cycle Data path): Hardwired and micro programmed, Multi-cycle implementation Basic. Necessity of multi-cycle implementation, Details of Control Signal Breaking the Instruction Execution into Clock Cycles, Exceptions	**2 weeks**	CLO2

	and Floating point Hardware		
Final Exam: 50 Marks **Group-A (20 Marks)**			
4	The control unit (Single cycle Data path) : hardwired and micro programmed, Logic Design Convention, Clocking Methodology, Data path Basic, State Element of Data path Building a single Data path (R-Type, I-Type and J-type Instructions) Designing the main control unit (with control signals effect), Operation of the data path	**2 week**	CLO2
5	The control unit (Multiple cycle Data path): Hardwired and micro programmed, Multi-cycle implementation Basic. Necessity of multi-cycle implementation, Details of Control Signal Breaking the Instruction Execution into Clock Cycles, Exceptions	**2 weeks**	CLO2
Group-B (30 Marks)			
6	Pipelining: Overview of Pipelining, Pipelining Hazard, Pipeline Data Path (R-Type, I-Type and J-type Instructions) Pipeline Control, Exception, Exception Handling	**1 week**	CLO3

7	Memory organization: Introduction to memory, memory Hierarchy, Basic on Cache, Accessing a cache, Handling cache miss and writes, Flexible placement of cache block	**2 weeks**	CLO3
8	I/O systems, channels, interrupts, DMA, I/O Devices basic, Disk Storage and dependability, RAID, Buses and other connections between processors, Memory and I/O Devices, Interfacing I/O devices with processors. Introduction to virtual Memory, Virtual to physical, Handling page Fault TLB, Integrating Virtual Memory, TLBs and Caches	**2 week**	CLO3